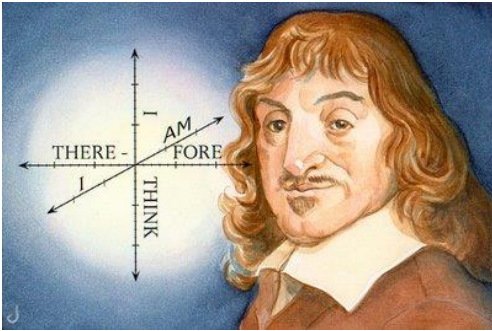


Definition

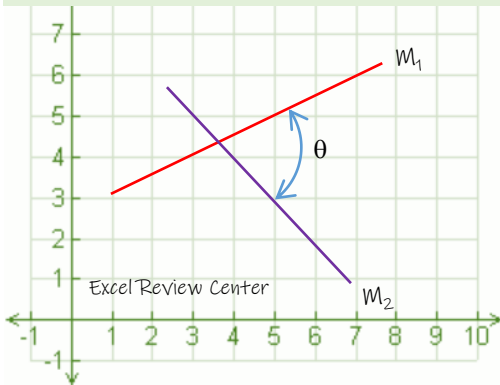
Analytic geometry deals with geometric problems using coordinates system thereby converting it into algebraic problems.
Excel Review Center



Rene Descartes (1596 – 1650, Cartesius in Latin language) is regarded as the founder of analytic geometry by introducing coordinates system in 1637.

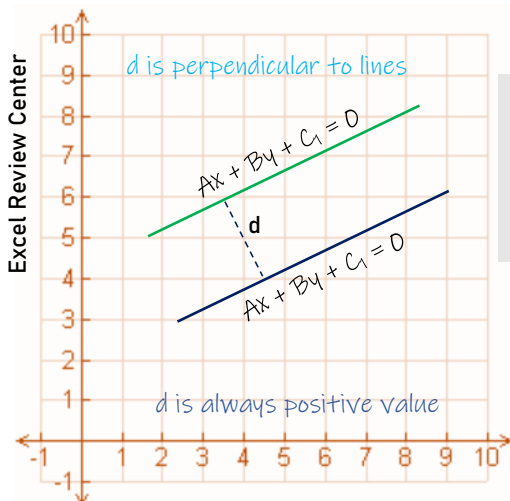
Angle between two Lines

$$\theta = \tan^{-1} \left(\frac{m_2 - m_1}{1 + m_2 m_1} \right)$$



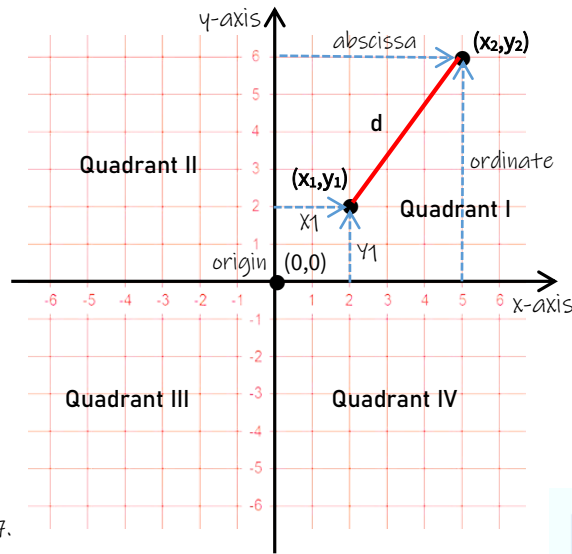
Angle between two Parallel Lines

$$d = \frac{C_1 - C_2}{\pm \sqrt{A^2 + B^2}}$$



Rectangular Coordinates System

Also known as Cartesian Coordinates System



Point O is the origin and has coordinates (0,0). The abscissa or x-coordinate is always measured from the y-axis while the ordinate or y-coordinate is always measured from the x-axis. The rectangular coordinates system is divided into four quadrants as shown.

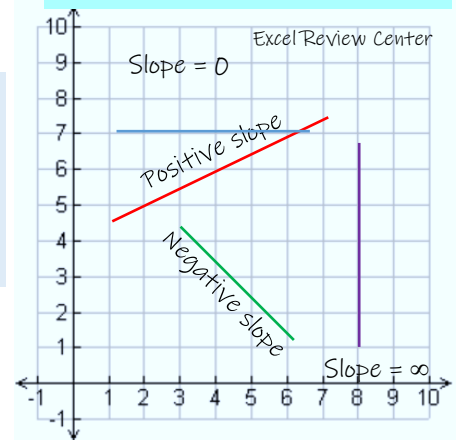
Distance Formula

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

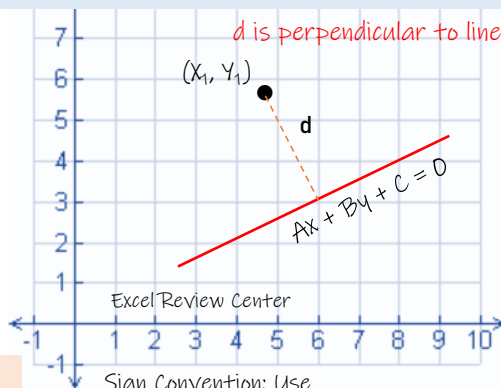
Slope of a Line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



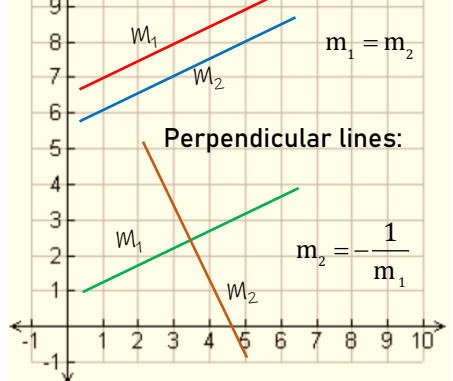
Distance between a Point and a Line

$$d = \frac{Ax_1 + By_1 + C}{\pm \sqrt{A^2 + B^2}}$$

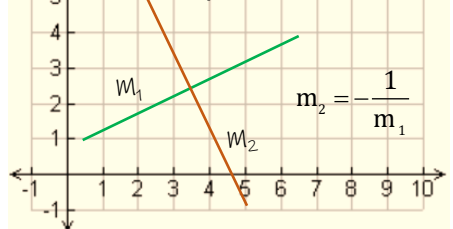


Sign Convention: Use
+ if B is positive and the point is above the line or to the right of the line
+ if B is negative and the point is below the line or to the left of the line
- If otherwise

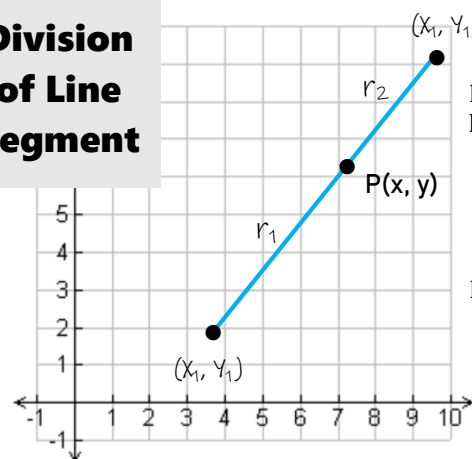
Parallel lines:



Perpendicular lines:



Division of Line Segment



Let r_1 and r_2 be the corresponding ratio of its length to the total distance between two points.

$$x = \frac{(x_1 r_2) + (x_2 r_1)}{r_1 + r_2}$$

$$y = \frac{(y_1 r_2) + (y_2 r_1)}{r_1 + r_2}$$

If P is at midpoint of two points.

$$x = \frac{x_1 + x_2}{2}$$

$$y = \frac{y_1 + y_2}{2}$$